



ARISTOTELIS BALLAS

Aristotelis Ballas is a Postdoctoral AI & ML researcher at Harokopio University of Athens, where he also completed his Ph.D. on representation learning algorithms for out-of-distribution generalization under Prof. Christos Diou. His research focuses on developing methods for learning robust representations that generalize beyond their training distribution, with applications spanning computer vision and biosignal analysis. His work has appeared in venues such as CVPR (spotlight) and IEEE Transactions Journals. He also holds a diploma in Electrical and Computer Engineering from the National Technical University of Athens.

PyTorch

Tensorflow

CI/CD

LaTeX

EDUCATION

02/26 – CURRENT

Harokopio University of Athens, Greece

11/21 – 12/25

Harokopio University of Athens, Greece

10/14 – 05/21

National Technical University of Athens, Greece

09/11 – 06/14

2nd High School of Glyka Nera, Greece

WORK EXPERIENCE

06/21 – CURRENT

Harokopio University of Athens, Greece

05/17 – 08/21

ALMA Bank, USA

POSTDOC IN MACHINE & DEEP LEARNING

Title: *Learning Robust Representations for OOD Generalization via Loss Landscape Analysis*

Supervisor: Prof. Christos Diou

Research Interests: *OOD Generalization, Representation Learning, Biosignal Classification, AI in Healthcare*

PHD IN MACHINE & DEEP LEARNING

Dissertation: *Representation Learning Algorithms for Out-of-Distribution Generalization*

Supervisor: Prof. Christos Diou, Grade: 10/10

DIPLOMA IN ELECTRICAL AND COMPUTER ENGINEERING

Integrated M.Sc degree in ECE (EQF Level 7)

Diploma Thesis: *Ωto_abR: A WebApp for the Visualization and Analysis of Click-Evoked Auditory Brainstem Responses*

Diploma Grade: 7.25/10.0

SCHOOL-LEAVING CERTIFICATE, 'APOLYTYRION'

Overall Grade: 19.3 / 20.0

AI & ML RESEARCHER

AI and ML researcher for the EU funded projects of REBECCA, RELEVIMUM and MELIORA.

- AIRIS: WP2 Leader - Mechanism-informed Multimodal Generative AI models. Research towards methods, tools and models for mechanism-constrained multimodal GenAI.
- REBECCA: Developed and integrated Early-Warning-System and Synthetic Data Generator for Counterfactual Interventions
- RELEVIMUM: Developed, implemented and integrated: (i) CNN-LSTM bite-recognition model from 1D smartwatch signals, (ii) VLM for food recognition from images. Managed CI/CD, backup and overall dev and prod system for the project, along with multiple rounds of integration testing.
- MELIORA: Developed LLM-based components for a virtual coach delivering behavioural change interventions for cancer risk reduction, along with VLM-based food-logging system.

SYSTEMS ENGINEER

Provided remote IT consultation. Responsible for design, configuration and development of the domain's system and network servers along with the fine-tuning and monitoring of the infrastructure.

WORK EXPERIENCE	05/16 – 04/21 ALMA Bank, USA	HELPDESK ANALYST Provided remote front-line primary technical support to end-users on various technical issues and problems relating to hardware, software and peripherals.
OTHER PROJECTS	2023 Harokopio University of Athens, Greece	HIGH-PERFORMANCE COMPUTING CLUSTER SETUP Core member of the team that was tasked to setup and administer HUA's HPC cluster. Responsibilities included: Setup of SLURM workload manager, setup of Ubuntu GPU servers, configuration of all NVIDIA GPU cards and drivers, including documentation.
TEACHING EXPERIENCE	2024 - CURRENT Harokopio University of Athens, Greece	DEEP LEARNING Postgraduate course (TA) - Computer Science M.Sc
	2023 - 2024 Harokopio University of Athens, Greece	ARTIFICIAL INTELLIGENCE AND ITS APPLICATIONS ON THE INTERNET OF THINGS Postgraduate course (TA) - Computer Science M.Sc
	2022 - 2025 Harokopio University of Athens, Greece	DATA MANAGEMENT II Postgraduate course - Applied Informatics M.Sc
ACADEMIC ACTIVITIES	TUTORIALS	TUTORIAL ON DOMAIN GENERALIZATION Presented in IEEE BDS 2023, this short tutorial was designed to introduce anyone interested to the problem of Domain Generalization and assist whoever wants to get into the field, navigate the literature. Website: https://bds-dgtutorial.github.io/
	JOURNAL REVIEWER	IEEE TRANSACTIONS ON MULTIMEDIA INTERNATIONAL JOURNAL OF COMPUTER VISION PATTERN RECOGNITION NPJ DIGITAL MEDICINE NEURAL NETWORKS SIGNAL, IMAGE AND VIDEO PROCESSING
	CONFERENCE REVIEWER	IEEE ENGINEERING IN MEDICINE AND BIOLOGY ACM INTERNATIONAL CONFERENCE ON MULTIMEDIA RETRIEVAL NEURIPS

SKILLS + INTERESTS

Deep learning	Libraries	IT Skills	Soft Skills	Interests
PyTorch Tensorflow Keras TinyML Pytorch-Mobile	Numpy Scipy Scikit-Learn Pandas OpenCV Matplotlib	Python, R, Matlab Docker, Jenkins, Git MongoDB, SQL, InfluxDB Windows, Linux SLURM LaTex, Wireshark, VMware	Multidisciplinary thinking Passion for learning Teamwork Able to work independently Attention to detail Project management	Cliff Diving Cinema enthusiast Tutoring Reading Dancing

LANGUAGES

Greek	English
Native	Bilingual – C2

PUBLICATION LIST	2025	A. Ballas and C. Diou. "Gradient-Guided Annealing for Domain Generalization." In: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2025, pp. 20558–20568. (Spotlight)
	2024	A. Ballas, V. Papapanagiotou, and C. Diou. "Which Augmentation Should I Use? An Empirical Investigation of Augmentations for Self-Supervised Phonocardiogram Representation Learning." In: IEEE Access 12 (2024), pp. 193459–193472. A. Ballas and C. Diou. "CycleMix: Mixing Source Domains for Domain Generalization in Style-Dependent Data." In: Proceedings of the 13th Hellenic Conference on Artificial Intelligence. 2024, pp. 1–4. A. Ballas and C. Diou. "CycleMix: Mixing Source Domains for Domain Generalization in Style-Dependent Data." In: Proceedings of the 13th Hellenic Conference on Artificial Intelligence. 2024, pp. 1–4. A. Ballas and C. Diou. "Multiscale and Multilayer Contrastive Learning for Domain Generalization." In: IEEE Transactions on Artificial Intelligence 5.12 (2024), pp. 6253–6266. A. Ballas and C. Diou. "Towards Domain Generalization for ECG and EEG Classification: Algorithms and Benchmarks." In: IEEE Transactions on Emerging Topics in Computational Intelligence 8.1 (2024), pp. 44–54.
	2023	A. Ballas and C. Diou. "CNNs with multi-level attention for domain generalization." In: Proceedings of the 2023 ACM International Conference on Multimedia Retrieval. 2023, pp. 592–596. A. Ballas and C. Diou. "CNN Feature Map Augmentation for Single-Source Domain Generalization." In: 2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService). 2023, pp. 127–131.
	2022	A. Ballas, V. Papapanagiotou, A. Delopoulos, and C. Diou. "Listen2YourHeart: A Self-Supervised Approach for Detecting Murmur in Heart-Beat Sounds." In: 2022 Computing in Cardiology (CinC). Vol. 498. 2022, pp. 1–4. A. Ballas and C. Diou. "A Domain Generalization Approach for Out-Of-Distribution 12-lead ECG Classification with Convolutional Neural Networks." In: 2022 IEEE Eighth International Conference on Big Data Computing Service and Applications (BigDataService). 2022, pp. 9–13. A. Ballas and C. Diou. "Multi-layer Representation Learning for Robust OOD Image Classification." In: Proceedings of the 12th Hellenic Conference on Artificial Intelligence. SETN '22. Corfu, Greece: Association for Computing Machinery, 2022.
	2021	Ballas, A., & Katrakazas, P. (2021). <i>ωto_abr</i> : A web application for the visualization and analysis of click-evoked auditory brainstem responses. <i>Digital</i> , 1(4), 188-197.